



8539 - Probing the distant universe through GRB afterglow spectroscopy

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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Name	Institution
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OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
Bright, $z < 7.3$				
	4	NIRSpec/FS high-res w ith F070LP + Prism	NIRSpec Fixed Slit Spectroscopy	(1) GRB1
Bright, $z > 7.3$				
	8	NIRSpec/FS high-res w ith F100LP + Prism	NIRSpec Fixed Slit Spectroscopy	(1) GRB1
Faint, $z < 7.3$				
	9	NIRSpec/FS medium-r es with F070LP + Pris m	NIRSpec Fixed Slit Spectroscopy	(1) GRB1
Faint, $z > 7.3$				
	10	NIRSpec/FS medium-r es with F100LP + Pris m	NIRSpec Fixed Slit Spectroscopy	(1) GRB1

ABSTRACT

Gamma-ray bursts are bright enough to be seen to very great distances and their afterglows can provide redshifts and positions for their host galaxies, as well as detailed characterisation of the host ISM and surrounding IGM, even for very faint hosts that are otherwise undetectable. Thus GRBs, despite their small numbers, offer a unique and powerful tracer of early star formation and the whole galaxy population through the era of reionization. Efforts to identify high- z GRBs have been rewarded with the discoveries of several with spectroscopic redshifts $z=6.5-8.5$. However, it remains the case that some good candidate high- z GRBs cannot be followed up quickly or deeply enough with ground-based IR spectroscopy, and indeed for others the Ly-alpha break may fall in regions of the IR spectrum difficult to access from the ground. GRB 090429B is an example, which had a photo- z of 9.4, but for which spectroscopy was curtailed due to bad weather. We propose target of opportunity spectroscopy of the afterglow of a GRB with good photometric or low-S/N spectroscopic evidence of $z>6.5$. The sensitivity of JWST is such that it can constrain host HI column, IGM neutral fraction, metal abundances, and dust in the high- z host galaxy, even several days post-burst. The recent launch of the EP and SVOM satellites will enhance the rate of discovery of high- z events, making this program urgent and timely. Thus we will obtain unique information about a typical high- z galaxy and contribute to building samples which will constrain population properties of primordial galaxies across the luminosity function, the average escape fraction of ionizing radiation, and the evolution of the IGM.

OBSERVING DESCRIPTION

The intention of this proposal is to obtain JWST spectroscopy of a GRB afterglow found during cycle 4, for which there is strong indication of $z > 6.5$, based on ground photometry or low S/N and/or low resolution spectroscopy, for which JWST observations even ~ 3 -10 days post-burst would provide significantly superior data. Given how little we know about very faint galaxies and the IGM in the era of reionization, important results may come from a single spectrum, but the program will also help build up samples which will provide stronger statistical constraints on e.g. the average escape fraction, the typical abundances, as well as targets for future host searches.

The science goals themselves depend on the nature of the GRB, in particular the afterglow brightness and also hydrogen column density and metallicity. In the case of a bright afterglow it will be possible to measure the Lyman alpha damping wing well, and hence decompose the host neutral hydrogen column and IGM neutral fraction proximate to the burst. If the host column and metallicity are sufficiently high, then we will constrain the abundances based on the least saturated lines, and these will also pin the precise systemic redshift, and hence improve the precision of the Lyman-alpha damping wing analysis.

A shorter prism observation has the capability of making a sensitive search for dust on the line of sight in the host galaxy, including constraining the dust law, which may provide remarkable insight into early dust formation (or lack of it, if dust is not seen).

The rapid triggering of this proposal is challenging, but the GRB community has long experience of working with STScI to make such time critical observations.

Proposal 8539 - Targets - Probing the distant universe through GRB afterglow spectroscopy

Generic Targets	#	Name	Criteria	Description
	(1)	GRB1	GRB afterglow with strong indications of $z>6.5$ for which JWST can improve substantially on ground data	

Proposal 8539 - Observation 4 - Probing the distant universe through GRB afterglow spectroscopy

Observation	Proposal 8539, Observation 4: NIRSpec/FS high-res with F070LP + Prism										Mon May 05 18:00:31 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(NIRSpec/FS high-res with F070LP + Prism (Obs 4)) Warning (Form): Response time < 14.0 Days is disruptive to the scheduling process. Only a limited number of disruptive ToOs will be accepted. (Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria		Description							
	(1)	GRB1	GRB afterglow with strong indications of z>6.5 for which JWST can improve substantially on ground data									
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	FULL	F140X	NRSRAPIDD6	3	1	1	171.788	147260	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A2				SUBS200A2			
Dithers	#	Primary Dither Positions							Sub-Pixel Pattern			
	1	5							NONE			
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140H/F070LP	S200A2	NRS	200	1	1	NONE	5	5	6239.892	227691
	2	G140H/F070LP	S200A2	NRS	200	1	2	NONE	5	5	6239.892	227691
	3	PRISM/CLEAR	S200A2	NRS	100	1	3	NONE	5	5	3123.892	227691

Proposal 8539 - Observation 4 - Probing the distant universe through GRB afterglow spectroscopy

Special Requirements	On Hold Awaiting identification of highz GRB target Target Of Opportunity Response Time 3 Days
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Proposal 8539 - Observation 8 - Probing the distant universe through GRB afterglow spectroscopy

Observation	Proposal 8539, Observation 8: NIRSpec/FS high-res with F100LP + Prism										Mon May 05 18:00:31 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(NIRSpec/FS high-res with F100LP + Prism (Obs 8)) Warning (Form): Response time < 14.0 Days is disruptive to the scheduling process. Only a limited number of disruptive ToOs will be accepted. (Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria		Description							
	(1)	GRB1	GRB afterglow with strong indications of z>6.5 for which JWST can improve substantially on ground data									
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	FULL	F140X	NRSRAPIDD6	3	1	1	171.788	147260	
Template	HFF Readout Mode				Slit			Subarray				
	false				S200A2			SUBS200A2				
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	5						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140H/F100LP	S200A2	NRS	200	1	1	NONE	5	5	6239.892	227691
	2	G140H/F100LP	S200A2	NRS	200	1	2	NONE	5	5	6239.892	227691
	3	PRISM/CLEAR	S200A2	NRS	100	1	3	NONE	5	5	3123.892	227691

Proposal 8539 - Observation 8 - Probing the distant universe through GRB afterglow spectroscopy

Special Requirements	On Hold Awaiting identification of high-z GRB target Target Of Opportunity Response Time 3 Days
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Proposal 8539 - Observation 9 - Probing the distant universe through GRB afterglow spectroscopy

Observation	Proposal 8539, Observation 9: NIRSpec/FS medium-res with F070LP + Prism										Mon May 05 18:00:31 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(NIRSpec/FS medium-res with F070LP + Prism (Obs 9)) Warning (Form): Response time < 14.0 Days is disruptive to the scheduling process. Only a limited number of disruptive ToOs will be accepted. (Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria		Description							
	(1)	GRB1	GRB afterglow with strong indications of z>6.5 for which JWST can improve substantially on ground data									
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	FULL	F140X	NRSRAPIDD6	3	1	1	171.788	147260	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A2				SUBS200A2			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	5						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140M/F070LP	S200A2	NRS	200	1	1	NONE	5	5	6239.892	227691
	2	G140M/F070LP	S200A2	NRS	200	1	2	NONE	5	5	6239.892	227691
	3	PRISM/CLEAR	S200A2	NRS	100	1	3	NONE	5	5	3123.892	227691

Proposal 8539 - Observation 9 - Probing the distant universe through GRB afterglow spectroscopy

Special Requirements	On Hold Awaiting identification of highz GRB target Target Of Opportunity Response Time 3 Days
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Proposal 8539 - Observation 10 - Probing the distant universe through GRB afterglow spectroscopy

Observation	Proposal 8539, Observation 10: NIRSpec/FS medium-res with F100LP + Prism										Mon May 05 18:00:31 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(NIRSpec/FS medium-res with F100LP + Prism (Obs 10)) Warning (Form): Response time < 14.0 Days is disruptive to the scheduling process. Only a limited number of disruptive ToOs will be accepted. (Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria		Description							
	(1)	GRB1	GRB afterglow with strong indications of z>6.5 for which JWST can improve substantially on ground data									
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	FULL	F140X	NRSRAPIDD6	3	1	1	171.788	147260	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A2				SUBS200A2			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	5						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140M/F100LP	S200A2	NRS	200	1	1	NONE	5	5	6239.892	227691
	2	G140M/F100LP	S200A2	NRS	200	1	2	NONE	5	5	6239.892	227691
	3	PRISM/CLEAR	S200A2	NRS	100	1	3	NONE	5	5	3123.892	227691

Proposal 8539 - Observation 10 - Probing the distant universe through GRB afterglow spectroscopy

Special Requirements	On Hold Awaiting identification of highz GRB target Target Of Opportunity Response Time 3 Days
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